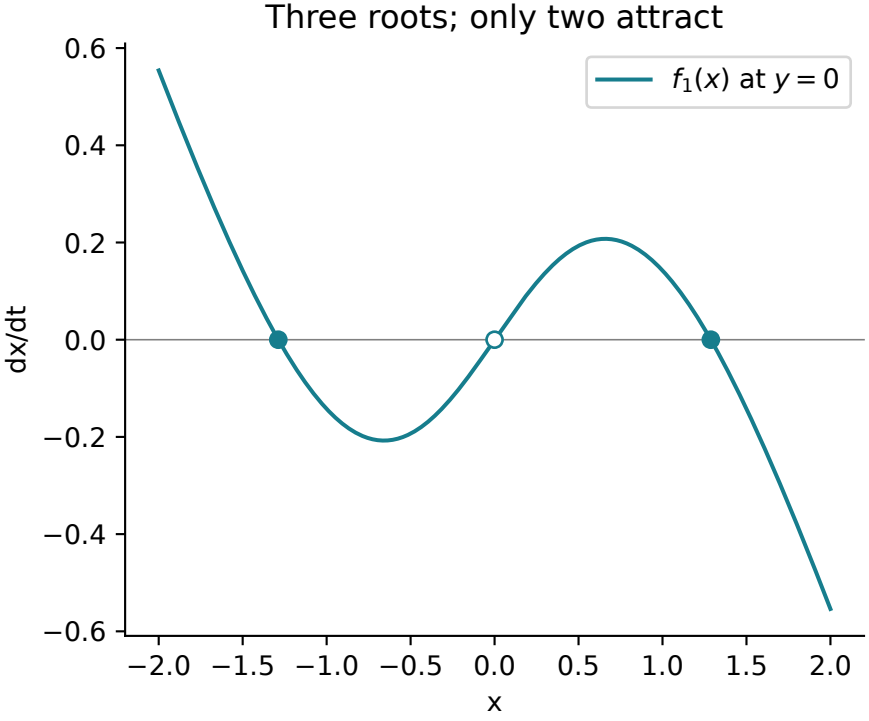
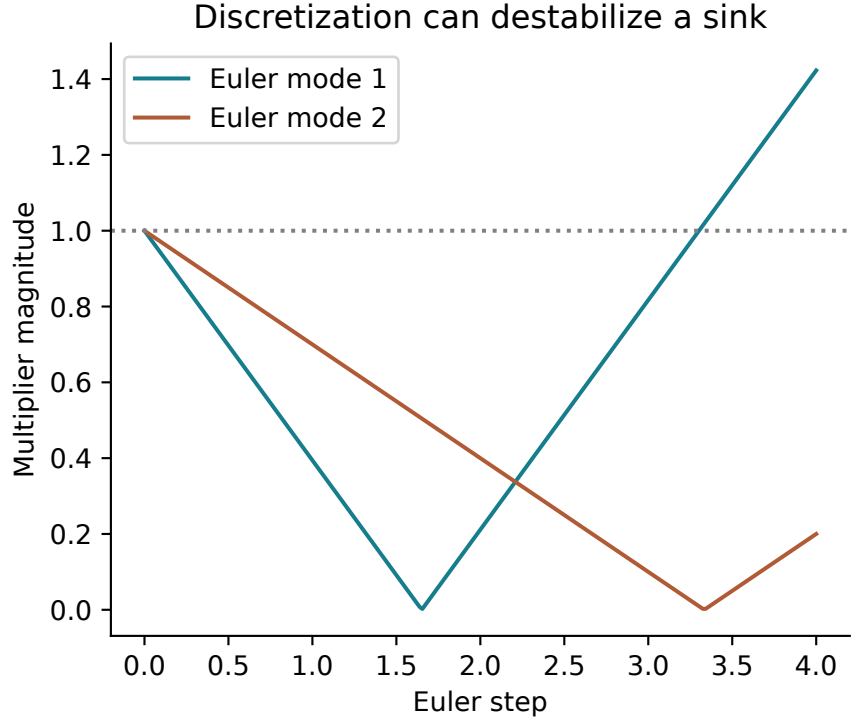

Three roots; only two attract
$f_1(x)$ at $y = 0$
dx/dt
x
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0


Discretization can destabilize a sink
Euler mode 1
Euler mode 2
Multiplier magnitude
Euler step
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0